Weather Radar System (Mod 6/2049) – General Data

1. General

The RDR 2000 weather radar (Radio Detecting and Ranging) system gives the pilot a selectable horizontal display of thunderstorms or high density precipitation in front of the aircraft. When a pulse intercepts a target, the energy is reflected as an echo, or return signal, back to the antenna. From the antenna, the returned signals are transferred to the receiver/transmitter unit. The returned signals, are then displayed.

The radar system can also be used for ground mapping (GMAP). In GMAP mode, prominent landmarks are displayed which enable the pilot to identify coastlines, hilly or mountainous regions, cities, or in some cases even large structures.

For aircraft incorporating customer Standard Order Option S.O.O. 6388 there is no Weather Radar System installed on the aircraft.

For aircraft with S.O.O. 6388 incorporated, the Weather Radar System and associated systems are installed by incorporating customer Standard Order Option S.O.O. 6383.

2. Description

A. WX RADAR Transceiver (RDR 2000)

The RDR 2000 installation consists of a 4kW radar transmitter, configuration module, a receiver, and an integral processing unit, all installed in the nose radome. The processing unit receives pitch and roll signals from the ADAHRS via ARINC 429 and processes this data to stabilize the 12 inch flat panel radar antenna. The sensor unit transmits a beam of pulsed microwave energy. When a pulse reaches visible moisture, the energy is reflected back to the antenna. The return signals are processed and sent to the Modular Avionics Unit (MAU). The weather radar system is connected to the air/ground monitor system, and is automatically switched to STANDBY mode when the aircraft is on the ground. The configuration module stores the configuration and calibration data for the RDR 2000 system.

The antenna pedestal elevation assembly permits the radar beam to be tilted up or down 15 degrees. Tilt may be controlled manually on the multi-function controller or may be determined automatically. In the case of automatic tilt, the manual tilt control provides a trim capability around the automatically set position. Stabilization capability of up to 30 degrees in pitch and roll is possible. Combined antenna stabilization plus tilt angle is functional over a range of ±30 degrees. The azimuth motor drives the antenna up to 50 degrees left and right for a total maximum scan angle of 100 degrees, although in practice a 90 degree image is scanned. The scan rate is 25 degrees per second, providing 16.7 “looks per minute”.

The system displays storm intensity levels in colour against a black background. Rainfall intensity is presented in four colours: green, yellow, red, and magenta, with magenta representing rainfall of the highest intensity. GMAP mode uses green, yellow and magenta colours to display the different reflectivities. Magenta represents the most highly reflective target. Separate sensitivity timing control (STC) compensation is used for weather and GMAP modes.

Weather radar information can be displayed as overlays on the PFDs and the navigation map display of the MFD. The PFD weather radar overlay can be assessed by pressing the soft key on the side of the HSI display. The soft key identifier OVRLY then appears in white. Pressing the OVRLY soft key displays the overlay selection menu. Selecting WX RDR will enable the weather radar overlay to be displayed on the HSI. There is also an OFF section to remove the overlay. The controls for the weather radar are located on the MF Controller. A WX RADAR dual concentric knob control with a push select function is used to control the mode and the tilt, gain settings.

The outer control is a four position rotary knob with the position OFF/STBY/TEST/WX. The inner control is rotary click control knob and is pressed to modify the tilt or gain setting.

Weather radar annunciations for ALERT, MODE and TILT are located on the left side of the HSI. The ALERT annunciations are TX ON GND in amber when WX is selected on the MF Controller and the aircraft is on the ground. TGT ALRT is given in amber when there are potentially hazardous targets directly in front of the aircraft that are outside of the selected range. Longer ranges should then be selected to view the questionable target. TGT is given in white when WX is selected and the aircraft is in the air. The MODE annunciation is that set by the WX RADAR outer control knob. The TILT annunciation value is a three digit number preceded by an arrow, up for positive value and down for negative value. Faults are annunciated WX FAULT in white on the right lower part of the weather radar overlay and failures are annunciated WX FAIL in amber. The Avionics window of the systems MFD contains WX/LX/TAWS setup pages. The WX setup tab is selected via the page menu of the multi-functional window. The RDR 2000 Weather Radar, when installed in combination with the Honeywell Primus Apex® integrated avionics suite, is not capable of displaying vertical depiction of weather. Refer to the Honeywell Primus Apex® Integrated Avionics System for the Viking Twin Otter Series 400 – Pilot Guide for complete information on the description and operation of the weather radar system.

The weather radar antenna and configuration module is mounted on the nose assembly radome bulkhead at FS-6.

28 Volt DC power is provided from the left bus through a 5 amp WX RADAR circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

The RDR 2000 weather radar system interfaces to the following aircraft systems:

1) Modular Avionics Unit (MAU) (refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data).

2) Air Data Attitude and Heading Reference System (ADAHRS) (refer to 34-23-00, Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) – General Data).

3. Adjustment/Test

WARNING **TESTS INVOLVING RADIATION OF RADIO FREQUENCY ENERGY BY THE RADAR ANTENNA MUST NOT BE MADE IN THE VICINITY OF REFUELLING OPERATIONS, OR WHEN PERSONNEL ARE STANDING DIRECTLY IN FRONT OF THE RADAR ANTENNA.**

RADIO FREQUENCY ENERGY IS GENERATED IN ALL MODES EXCEPT STANDBY AND TEST. DO NOT ALLOW PERSONNEL WITH THE MPEL (MAXIMUM PERMISSIBLE EXPOSURE LEVEL) BOUNDARY BEING SCANNED BY THE ANTENNA WHEN THE SYSTEM IS TRANSMITTING.

A SAFE DISTANCE OF 15 FEET FROM THE ANTENNA RADOME IS RECOMMENDED.

A. WX RADAR Transceiver

1) Testing of WX RADAR Transceiver (RDR 2000)

a) Ensure the aircraft is outside, away from all hangers and personnel.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait several minutes for the aircraft electrical and avionics systems to initialize.

d) Ensure that WX RADAR (P3) circuit breaker on the avionics circuit breaker panel is closed.

e) Warm-up the weather radar by placing the unit in STBY for at least 60 seconds.

f) Set the weather radar control to TEST.

g) Adjust LH/RH PFDs range to 100 Nm. Ensure the test pattern shows the green, yellow, red and magenta colour bands on the LH/RH PFDs and MFD.

h) In the MFD WX tab page on the MFD, press NEXT to show the ANT STABILIZE and WX ROLL TRIM adjust page.

i) Ensure ANT STABILIZE is selected.

j) Gain access to the forward avionics shelves through the forward baggage compartment and locate the ADAHRS (refer to 34-23-00, Air Data, Attitude and Heading Reference System (ADAHRS) (KSG 7200) – General Data).

k) Loosen retaining screw clamp from the front lip of the ADAHRS mounting tray.

l) Tilt the ADAHRS to simulate a right hand turn. Visually confirm that the weather radar compensates by tilting to the left.

m) Tilt the ADAHRS to simulate a left hand turn. Visually confirm that the weather radar compensates by tilting to the right.

n) Tilt the ADAHRS to simulate nose down. Visually confirm that the weather radar compensates by tilting up.

o) Tilt the ADAHRS to simulate nose up. Visually confirm that the weather radar compensates by tilting down.

p) Return ADAHRS unit to its mounting tray. Tighten screw clamp at the front lip of the ADAHRS mounting tray.

q) Enable the transmit on ground function by selecting the WX tab in the WX/LX/TAWS sensors page.

r) Align the aircraft such that a target that local cloud conditions are viewed on the weather radar display.

s) Place the radar in WX mode.

t) Turn the multi-function control (MFC) weather radar TILT/GAIN control. Ensure that the GAIN on the display changes.

u) Press the multi-function control (MFC) weather radar TILT/GAIN control. Ensure the TILT value changes from +15 to –15 degrees when the control is turned fully clockwise and than fully counterclockwise.

v) Enable the GMAP function by selecting the WX tab in the WX/LX TAWS sensors page. Ensure the GMAP displays a picture of the local conditions.

w) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Configure Module Setup (CM 2000) (Pre Mod 6/2200)

a) Connect the KPA–900 configuration module programming kit in conjunction with a personal computer (PC).

b) Ensure that the 9V radio battery in the KPA–900 is fully charged and that the KPA–900 is connected to your PC printer port.

c) The DOS program will only work with Windows up to 98 version, it will not work with later windows software.

d) Boot up your PC in DOS and work solely in DOS using the 3.5” floppy.

e) When you have entered the configuration page, go to Hardware, and select write. This puts default settings into the configuration module.

f) Go to the Memory and select View. Select each item and ensure a check mark appears to the left of the item as shown in Table 1 — CM 2000 Configuration Module Settings.

g) When you have selected all items as shown in Table 1 — CM 2000 Configuration Module Settings, depress the ESC button on PC.

h) Go to Hardware and select write. When the program has finished writing and verified itself, select Exit.

i) Remove the KPA–900 configuration module programming kit from the CM 2000 configuration module and from the PC.

3) Configure Module Setup (CM 2000) (Post Mod 6/2200)

a) Connect the KPA–900 configuration module programming kit in conjunction with a personal computer (PC) to the printer port.

b) Wait at least 2 minutes for the computer to reconfigure its own printer port. If self configuration is not complete before the KPA–900 maintenance program is started, the program will not detect a low battery status correctly, and the CM 2000 will not program correctly. (When program starts it checks the battery).

c) The Windows program will only work with Windows 95 up to Windows XP, it has not been tested for Windows VISTA or higher.

d) Start the KPA–900 software.

e) Select 'Unit Type' – 'ART 2000'.

f) Ensure the correct Radial Button is selected to the left of each item, as shown in Table 1 — CM 2000 Configuration Module Settings.

g) Select 'Tools – Write to CM 2000'.

h) Verify the activity log.

i) Select 'File – Save Log File'. (Optional).

j) Close program.

k) Remove the CM 2000 configuration module from the KPA–900 configuration module programming kit, and the KPA–900 kit from the PC.

Table 1 CM 2000 Configuration Module Settings

CM 2000 Configuration Module Settings	
Configuration Selection	**Setting**
Gyro Source	429 Bus
Analog Gyro Type	Gyro Source set to 429 bus will cause the Analog Gyro Type option to be ignored
Gyro Compensation	Gyro Source set to 429 bus will cause the Analog Gyro Type option to be ignored
Mechanical Scan Limit	–
Connector Position	Bottom
Attitude Source	–

Table 1 CM 2000 Configuration Module Settings (continued)

CM 2000 Configuration Module Settings	
Configuration Selection	**Setting**
Desired Antenna Sweep	100 degrees
Wx Gain	–
Map Gain	Gain Change Accepted
EFIS Strut Switch	Override Disallowed
ARL	–
Auto Step Scan	–
Sector Scan	–
Auto Tilt	–
Target Alert	Enabled

B. Electrical Bonding Test – Weather Radar Antenna

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to the Weather Radar Antenna which is mounted on the forward bulkhead inside the radome nose assembly.

3) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the Weather Radar mounting screw. Place the second probe on the nearest ground stud. The resistance between the two points must be < 3.0 milliohms unless stated otherwise.

4) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

5) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

6) Reattach radome nose assembly and tighten retaining screws. Refer to 53-10-02, Forward Baggage Compartment – General Data.

4. Removal/Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – ALL RACK MOUNT / REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LRU MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. WX RADAR Transceiver/Antenna (RDR 2000)

1) Removal

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Pull the WX RADAR (P3) circuit breaker on the avionics circuit breaker panel and fit safety clips.

c) Gain access to the transceiver which is mounted on the forward bulkhead behind the radome nose assembly.

d) Disconnect the electrical connector on the transceiver unit.

e) Remove the four screws that secure the transceiver unit to the radome bulkhead.

f) Remove unit from aircraft.

Note Stow connectors of the removed components.

2) Installation

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the WX RADAR (P3) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clips.

c) Gain access to the weather radar transceiver, mounted on the forward bulkhead behind the radome nose assembly.

d) Align the weather radar transceiver mounting holes to the radome bulkhead. Ensure correct vertically polarized orientation, connector faces down.

e) Reconnect the electrical connector to the radar transceiver unit. Ensure connections are secure, and sound.

f) Remove the screw that secures the cover from the Data Interrupt Program (DIP) switches, located to the right of the electrical connector.

g) Ensure that the Data Interrupt Program (DIP) switch settings are in the correct position (refer to Table 2 — Data Interrupt Program (DIP) Switch Positions).

Note A setting is in the ON position when the switch is facing the center of the unit.

Table 2 Data Interrupt Program (DIP) Switch Positions

DIP Switch Number	Switch Position
1	ON
2	ON
3	OFF
4	OFF
5	ON
6	OFF
7	ON
8	OFF

h) Install the screw that secures the Data Interrupt Program (DIP) switch cover to the transceiver unit.

i) Remove safety clips and close WX RADAR (P3) circuit breaker on the avionics circuit breaker panel.

j) Test the weather radar system as per Para 3.A.1).

B. Configuration Module (CM 2000)

1) Removal

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Pull the WX RADAR (P3) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Gain access to the configuration module which is mounted on the forward bulkhead inside the radome nose assembly.

d) Remove the two screws that secure the configuration module to the radome bulkhead.

e) Remove configuration module from aircraft.

Note Stow connectors of the removed components.

2) Installation

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the WX RADAR (P3) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clip.

c) Gain access to the configuration module which is mounted on the forward bulkhead inside the radome nose assembly.

d) Align the configuration module mounting holes to the radome bulkhead. Ensure correct orientation.

e) Reconnect the electrical connector to the configuration module. Ensure connections are secure, and sound.

f) Remove safety clip and close WX RADAR (P3) circuit breaker on the avionics circuit breaker panel.

g) Verify the configuration module setup as per Para 3.A.2).

5. Troubleshooting

A. WX RADAR Transceiver (RDR 2000)

1) If the WX RADAR is deemed inoperative check the following:

a) Ensure WX RADAR (P3) circuit breaker is closed.

b) Visually inspect WX RADAR connector for damaged wires or contacts.

c) Visually inspect WX RADAR antenna for corrosion and contaminants.

d) Ensure that the configuration module (CM 2000) and WX RADAR connector is properly mated and secure.

e) Inspect antenna external surfaces for damage.

f) Check that the antenna is properly mounted and screws are tight.

g) Confirm configuration module settings as per Table 1 — CM 2000 Configuration Module Settings.

h) If troubleshooting at the wire level is required refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

i) If repair or service is required, send to appropriate rated avionics repair facility.